

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

MATHEMATICS

9740/01

Paper 1

15 September 2009

3 hours

Additional Materials: Answer Paper
 Cover Page
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **7** printed pages and **1** blank page.



- 1 The price per guest for a buffet lunch at a hotel is shown in the following table.

	Price per guest
Children aged 12 and below	\$ a
Senior citizens aged 65 and above	\$ b
Others	\$ c

If a guest has lunch at the hotel on his or her birthday, the guest will receive a percentage discount equal to his or her age. For example, a guest who has lunch at the hotel on his 40th birthday will pay only 60% of the price. This offer is valid only for that guest, with the accompanying guests paying the usual price.

On a certain day, three groups having lunch at the hotel had a member each whose birthday was on that day.

- The first group, which consisted of two children below 12 years old, a grandparent celebrating her 65th birthday, and three adults below 65 years old paid \$200.88.
- The second group, which consisted of five children below 12 years old, two grandparents above 65 years old, and five adults below 65 years old (one of whom was celebrating his 35th birthday) paid \$404.52.
- The third group, which consisted of a child celebrating her 10th birthday, a grandparent above 65 years old, and two adults below 65 years old paid \$153.12.

Find the values of a , b and c . [4]

- 2 (i) State the derivative of $e^{\sin \frac{1}{2}x}$. [1]

- (ii) Find $\int e^{\sin \frac{1}{2}x} \sin x \, dx$. [4]

- 3 The complex number z is given by

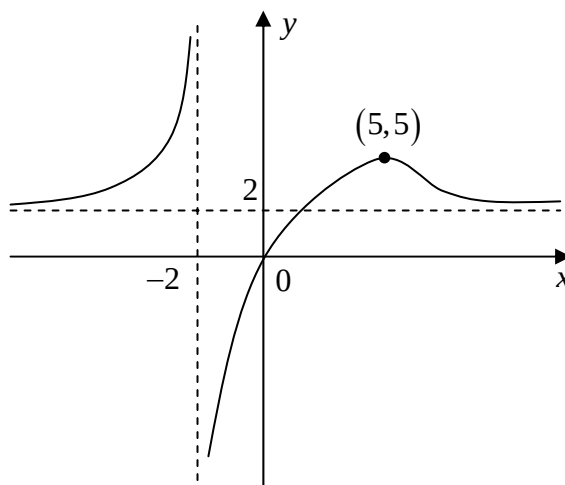
$$z = r e^{i\alpha}, \text{ where } r > 0 \text{ and } 0 < \alpha < \frac{\pi}{2}.$$

Given that z satisfies the equation

$$z^5 - |z|^4 z^* = 0,$$

- (i) show that z^6 is real, [2]
- (ii) find the exact value of α . [3]

- 4 The diagram shows the graph of $y = f(x)$. The curve has a maximum point $(5, 5)$ and it passes through the origin. The equations of the asymptotes are $x = -2$ and $y = 2$.



On separate diagrams, sketch the graphs of

(i) $y^2 = f(x)$, [3]

(ii) $y = \frac{1}{f(x)}$, [3]

indicating clearly the equations of any asymptotes and the coordinates of any stationary points and intersections with the axes.

- 5 A sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = \frac{1}{6}$ and

$$u_n = u_{n-1} + \frac{n2^{n-1}}{(n+1)(n+2)}, \quad \text{for all } n \geq 2.$$

Use the method of mathematical induction to prove that

$$u_n = \frac{2^n}{n+2} - \frac{1}{2}. \quad [6]$$

Deduce that $2u_n + 1 > 0$. [1]

- 6 Show that the differential equation

$$\frac{dy}{dx} = \pi y + \square(e^{2\pi x} - 3y^2)$$

may be reduced by the substitution $y = e^{\pi x} z$ to the equation

$$\frac{dz}{dx} = \square(1 - 3z^2).$$

Hence find y in terms of x , given that $y = \frac{1}{2}$ when $x = 0$. [7]

- 7 Find the expansion of $\left(\frac{1-x}{1+2x}\right)^{\frac{1}{3}}$ in ascending powers of x , up to and including the term in x^2 . State the range of values of x for which the expansion is valid. [5]

Without performing any calculations, explain why putting $x = \frac{1}{6}$ into the result gives a better approximation to $\sqrt[3]{5}$ than putting $x = -\frac{4}{11}$. [1]

Hence by putting $x = \frac{1}{6}$, find an approximation for $\sqrt[3]{5}$, expressing your answer as a fraction in its lowest terms. [2]

- 8 A certain type of bacteria is grown for research. The number of bacteria at the end of n days is modelled by u_n , where

$$u_{n+1} = u_n + 2n^2 - 5n,$$

and u_0 represents the initial number of bacteria.

By expressing u_1, u_2, u_3, u_4 in terms of u_0 , or otherwise, show that

$$u_n = u_0 + kn(n-1)(4n-17),$$

where k is a constant to be determined. [6]

Given that there are initially 10 bacteria, find the number of days it takes for the number of bacteria to first exceed 2 000. [2]

[The formula for $\sum_{r=1}^n r^2$ is $\frac{1}{6}n(n+1)(2n+1)$.]

- 9 Given that $y = \sin[\ln(1-3x)]$, prove that

$$(1-3x)^2 \frac{d^2y}{dx^2} - 3(1-3x) \frac{dy}{dx} + 9y = 0.$$

Hence find Maclaurin's series for y , up to and including the term in x^3 . [6]

Deduce an expansion for $\cos[\ln(1-3x)]$ up to and including the term in x^2 . [2]

- 10 (a) The function f is defined by

$$f : x \mapsto x - 2 \tan^{-1}(2x), \quad -\frac{\sqrt{3}}{2} < x < \frac{\sqrt{3}}{2}.$$

- (i) Show that $f(x)$ decreases as x increases. [3]
- (ii) Find the range of f . [1]
- (iii) Explain whether the composite function ff exists. [2]
- (iv) The curve C is obtained by reflecting the graph of f in the line $y = x$. Write down the equation of C . [1]

- (b) The function h is defined by

$$h : x \mapsto x^2 + 2, \quad x \in \mathbb{R}.$$

The function g is such that the composite function gh exists and

$$gh(x) = x^4 - 2x^2 + 5, \quad x \in \mathbb{R}.$$

Find an expression for $g(x)$. [2]

- 11** The equation of a closed curve is given by $(nx + y)^2 + 4(nx - y)^2 = 20$, where n is a constant. Find

- (i) the gradient of the curve at the point (x, y) , [2]
- (ii) the coordinates of the points on the curve at which the tangent is parallel to the y -axis, [4]
- (iii) the equations of the normal at the points where the curve cuts the x -axis. [4]

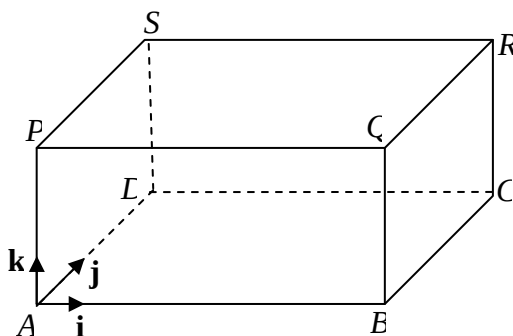
- 12** A geometric series G is given by

$$e - e^{1+\alpha} + e^{1+2\alpha} - e^{1+3\alpha} + \dots,$$

where α is a negative real number.

- (i) Show that G is convergent. [2]
- (ii) The sum to infinity of G is S . Given that the sum to infinity of the even numbered terms of G , i.e. the second, fourth, sixth, ... terms, is $-\frac{1}{3}S$, find the exact value of α . [4]
- (iii) In another series H , each term is the modulus of the corresponding term of G . Given that $\alpha = -0.03$, find the greatest value of n such that the product of the first n terms of H exceeds 15 000. [4]

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The diagram shows a cuboid with horizontal rectangular base $ABCD$, where $AB = 5$ units, $AD = 3$ units and $AP = 2$ units. The edges AP , BQ , CR and DS are vertical, and $PQRS$ is the top of the cuboid. The point A is taken as the origin and vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, each of length 1 unit, are taken along AB , AD , AP respectively.

- (i) Show that the line DQ has equation

$$\mathbf{r} = 3\mathbf{j} + \lambda(5\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}),$$

where λ is a parameter.

Find an equation of the line BS . [4]

- (ii) Hence calculate the acute angle between the lines DQ and BS , giving your answer to the nearest 0.1° . [3]

The point X lies on RC such that $RX = \frac{2}{3}RC$, and the point M is the midpoint of BD . The plane $\square$ contains the points R , B and M .

- (iii) Find, in scalar product form, an equation of the plane $\square$. [3]

- (iv) Find the exact value of the shortest distance from X to the plane $\square$. [3]